RAJALAKSHMI ENGINEERING COLLEGE

Narun Sabareesh K 250701469-2025-CSE

N2

Started on	Friday, 25 September 2026, 8:25 AM
State	Finished
Completed on	Friday, 25 September 2026, 8:49 AM
Time taken	23 mins 57 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>
int main()
{
    char a[100], b[100];
    int dp[100][100];
    scanf("%s", a);
    scanf("%s", b);
    int m = strlen(a);
    int n = strlen(b);
    for (int i = 0; i <= m; i++)
    {
        dp[i][0] = 0;
    }
    for (int j = 0; j <= n; j++)
    {
        dp[0][j] = 0;
    }
    for (int i = 1; i <= m; i++)
    {
        for (int j = 1; j <= n; j++)
        {
            if (a[i - 1] == b[j - 1])
            {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            }
            else
            {
                if (dp[i - 1][j] > dp[i][j - 1])
                {
                    dp[i][j] = dp[i - 1][j];
                }
                else
                {
                    dp[i][j] = dp[i][j - 1];
```

```
36                  }
37              }
38          }
39      }
40      printf("%d", dp[m][n]);
41      return 0;
42  }
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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